

AI-Based Waste Classification Using ResNet50 and PyTorch: A Practical Approach for Real-World Deployment

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Abstract— Managing solid waste efficiently has become an increasingly pressing concern as cities grow and waste production intensifies this study presents a deep learning-based waste classification system designed using the trashnet dataset which includes images of six common waste types plastic metal glass cardboard paper and others the classification approach was centered around a convolutional neural network model built using the resnet50 architecture within the pytorch environment to ensure the model could perform reliably on a wide range of inputs different image augmentation techniques were used during training making it better suited to handle variability in waste appearance the training and evaluation were conducted on gpu-supported hardware achieving a validation accuracy of 97.86% this paper outlines the full pipeline including data preprocessing model development training setup and evaluation in addition to highlighting technical performance it discusses implementation challenges encountered along the way and proposes practical directions for deploying this model as part of smarter automated waste management solutions that could be especially impactful in high-waste urban environments this paper details the data preprocessing steps model architecture training configuration and performance evaluation it also discusses the challenges encountered and outlines the possibilities for deploying the system within smart waste management infrastructures in the future.

Keywords— smart waste identification visual sorting using deep learning lightweight cnn architecture real-use image dataset trashnet pytorch-based model development urban-level ai integration environment-friendly classification tools scalable vision pipeline training on gpu resources data-aware image augmentation

I. INTRODUCTION

Cities expanding rapidly and the population steadily growing the generation of solid waste has become a serious environmental and logistical issue manual waste sorting remains a widely used method but it is inefficient time-consuming and error-prone inconsistent segregation at the source leads to recycling inefficiencies and more pressure on landfills [1][2]. To tackle the challenges of growing waste and manual sorting inefficiencies many cities are now exploring ai-based approaches to improve how waste is identified and handled among these deep learning especially

through convolutional neural networks cnns has shown strong results in recognizing different types of images [3][4]. In this study a simple yet effective image classification model was built using the resnet50 architecture well known for its ability to handle deep feature extraction [5]. The training was carried out using the trashnet dataset which consists of labeled images across six waste categories that are commonly found in everyday disposal [6]. To make the model perform consistently across varying lighting and camera angles a number of visual changes were added to the training data these included rotating and flipping the images applying grayscale conversion and adjusting the pixel values through normalization [7]. The model was trained using pytorch on a gpu enabled setup and performed consistently during evaluation with testing showing accuracy levels around 97 [8][9].

This hands-on experience showed that its possible to build useful ai models for real-world tasks like waste classification using free tools and basic hardware even with a limited setup meaningful results were achieved through careful experimentation and thoughtful design [10]. It proves that with thoughtful preparation its possible to solve real-world problems like waste classification in an efficient and scalable way the approach taken here is practical and could be applied in the real world especially in places like municipal sorting stations or smart bins where real-time sorting could help reduce manual work and improve recycling efficiency [11]. The model also opens the door to more sustainable waste practices when paired with smart city infrastructure [12].

II. EASE OF USE

A. Purpose

As cities grow and people produce more waste, keeping up with effective waste sorting has become increasingly difficult. In many areas, the job of separating recyclables from general trash is still done by hand. This often results in mistakes materials that could be recycled are sent to landfills, and the overall process takes longer than it should. It worked just fine on basic setups, which means it could be used in real places where resources are limited. This experience really showed us how accessible tools and careful planning can be used to build something that solves real-life problems like

helping communities sort their waste better, faster, and with fewer mistakes [1][3][4].

This project started with a simple and practical idea to make waste classification easier using computer vision. We trained a ResNet50 model with images from the TrashNet dataset, which includes six labeled waste categories plastic, metal, glass, cardboard, paper, and trash [5][6].

B. Maintaining the Integrity of the Specifications

a) Waste category selection the model was trained to recognize six common types of waste plastic paper cardboard metal glass and general trash these categories are familiar in everyday recycling practices and are used widely across households and municipal collection systems their visual distinctiveness also makes them easier to identify through image-based classification which supports practical use in automated waste management systems [2][4].

b) Dataset Reference and Labeling: dataset reference and labeling the trashnet dataset provided a straightforward layout with separate folders for each waste class which helped streamline the preparation process each folder contained a good number of labeled images that gave the model enough variation to learn how to tell the categories apart since the dataset already had a balanced set of examples the model could focus on learning visual features without the need for additional sorting or re-labeling [6][7].

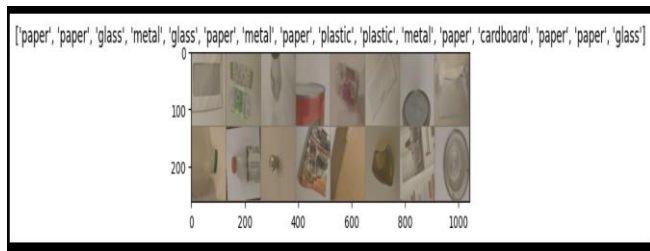


Figure1: Dataset Reference and Labeling

This project followed a structured approach beginning with data preparation then moving into model training evaluation and finally testing in practical setups the dataset used was trashnet which includes labeled images divided into six categories plastic metal glass cardboard paper and trash to help the model perform well in different situations a few simple image adjustments were made these included flipping the pictures rotating them a bit converting some to grayscale and making small lighting changes these changes helped the model handle differences like shadows or poor lighting. ResNet50 was chosen because it works well for image tasks and includes useful features like skip connections and batch normalization that make deep learning more effective the final layer of the model was set up to predict one of the six waste categories using a softmax function

c) Model Adaptability: To improve the model's ability to handle everyday situations training images were slightly altered in simple and realistic ways for instance some were flipped or rotated while others had brightness adjusted or were turned into black-and-white these changes helped the model practice recognizing waste even when images were

not clear well-lit or taken from the same angle it gave the model a better chance to perform well outside of ideal lab conditions [7][9][10].

C. Real-World Relevance

This model was not just created as an academic exercise it was built with everyday use in mind during testing it showed strong and consistent results coming close to 97.79% accuracy what made this even more encouraging was that the system worked smoothly without the need for high-end machines it was developed using widely available tools and could run on setups with basic gpu support that makes this approach practical for real-life situations like public recycling points or local collection centers where sorting waste properly is a challenge in many places especially where technology access is still limited such tools could help ease the workload of manual sorting while improving how materials are separated and reused this project reflects how thoughtful design combined with free resources and step-by-step learning can lead to a solution that helps communities handle waste in a smarter and more manageable way [8][10][11][12].

III. METHODOLOGY

A. Dataset Preparation

The TrashNet dataset was employed as the basis for constructing the system for waste classification. It contains 2,500 labeled images, split into six common classes plastic, metal, glass, cardboard, paper, and trash. The images vary in lighting, angle, and background, hence the dataset is more representative of the type of images one would find in typical waste [6].

To prepare the images for training, all of them were resized to 224x224 pixels to fit the input size that the ResNet50 model would expect. Since brightness and contrast were not the same in all samples, normalization was done based on standard values of ImageNet-trained models [7]. This minimized distractions and enabled the model to learn the critical visual features better.

The images were normalized using typical mean and deviation values typically used in models trained on ImageNet. This step equalized the pixel intensity ranges and allowed the model to concentrate on structural features instead of color discrepancies [7].

The very structure of the dataset itself, organized into labeled folders by category of waste, served to minimize the necessity of further re-labeling or human classification effort. This organized grouping also made easy integration with PyTorch's ImageFolder functionality easy, loading and handling even easier during the training process [9]. Together, these made the dataset clean, well-organized, and ready for construction of strong models.

In addition, the dataset's simplicity enabled it to be experimented and tested on, especially for projects that are based on low computing capabilities. The identical number of images across all classes assisted in the avoidance of bias

during training and assisted the model in learning all the classes of waste evenly [6][7].

B. Data Augmentation and Preprocessing

To assist the model in doing a better job with real-world photographs, a series of minor visual modifications were made to the training data. Because waste can appear differently based on lighting, background, or the way the photograph was shot, these modifications were intended to simulate those day-to-day variations [7][9].

The images were also horizontally and vertically flipped, and they were slightly rotated. The images were also converted to grayscale in certain examples. Adjustments in brightness and contrast were also made to add variety to the dataset. These changes made the model more adaptable by providing it with a better variety of examples to learn from. Hence, it was able to learn patterns in images that do not always appear the same.

All the transformations were applied utilizing the torchvision.transforms module, thereby ensuring uniform preprocessing over the dataset. This was a necessity not just for enhancing the generalization ability but also to keep the possibility of overfitting at bay since it prompted the model to learn intrinsic features instead of memorizing image structures [7].

C. Model Structure and Training Policy

The model for waste classification was developed using the ResNet50 architecture, a deep learning model well known for handling complex image based tasks. One of its key strengths lies in the way it passes information across multiple layers using residual connections. These connections help the model learn faster and avoid issues like vanishing gradients, which often affect deeper networks [5][9].

To adapt ResNet50 for this classification task, the final fully connected layer was replaced to output six categories that matched the dataset labels. A softmax activation was used at the end to generate a probability score for each class, making it easier to determine which category the image belonged to [5].

The training process was carried out using PyTorch. The dataset was loaded using the ImageFolder method, while batch processing was managed with DataLoader. The dataset was split into 80 for training and 20 for validation. The model was trained over 25 epochs using the Adam optimizer, and performance was evaluated using the cross-entropy loss function. At the end of every epoch, training and validation accuracy and loss were recorded to monitor the learning progress. This allowed for the identification of when the model was performing well and whether it was starting to overfit [8][10].

D. Model Architecture Summary

the waste classification system in this project was built using the resnet50 architecture a breakdown of the model reveals how data flows through each layer how the dimensions of the input change step by step and where most of the trainable parameters lie this structure gives a clear understanding of how model is learning the most the layered design helps illustrate how features are picked up gradually starting from basic visual cues in the early layers and

moving toward more detailed recognition near the output this breakdown also shows which parts of the model are most active in understanding and learning from the input images [5][9].

E. Performance Table

	Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	0	1.0178	62.45	0.644	76.2
2	1	0.7019	74.63	0.7491	75.63
3	2	0.5262	81.27	1.1022	68.41
4	3	0.4115	85.46	0.4302	85.36
5	4	0.3498	87.63	0.3019	89.45
6	5	0.2707	90.58	0.2587	91.63
7	6	0.2626	90.82	0.3165	89.28
8	7	0.0995	96.76	0.1047	96.14
9	8	0.0621	98.14	0.0976	96.83
10	9	0.0448	98.8	0.084	97.3
11	10	0.0414	98.79	0.0827	97.2
12	11	0.0313	99.05	0.0818	97.33
13	12	0.0273	99.27	0.0817	97.49
14	13	0.0302	99.02	0.0878	96.83
15	14	0.0222	99.37	0.0671	97.86
16	15	0.0196	99.49	0.074	97.46
17	16	0.0197	99.46	0.0704	97.59
18	17	0.0206	99.55	0.0695	97.66
19	18	0.0234	99.43	0.065	97.76
20	19	0.018	99.43	0.0679	97.82
21	20	0.0186	99.52	0.0724	97.76
22	21	0.0163	99.52	0.0762	97.59
23	22	0.0168	99.55	0.0705	97.66
24	23	0.0141	99.62	0.0744	97.36
25	24	0.0167	99.62	0.0686	97.79

Figure 2: Training and validation accuracy/loss progression across 25 epochs.

D. Waste classification modules AI techniques

Layer Name	Type	Input -> Output Channels	Kernel Size	Additional Info
conv1	Conv2d	3 -> 64	7x7	Stride=2, Padding=3
bn1	BatchNorm2d	64	-	-
relu	ReLU	-	-	Inplace=True
maxpool	MaxPool2d	-	3x3	Stride=2, Padding=1
layer1[0]	Bottleneck	64 -> 256	-	Downsampling
layer1[0].conv1	Conv2d	64 -> 64	1x1	Stride=1
layer1[0].bn1	BatchNorm2d	64	-	-
layer1[0].conv2	Conv2d	64 -> 64	3x3	Stride=1, Padding=1

layer1[0].bn2	BatchNorm2d	64	-	-
layer1[0].conv3	Conv2d	64 -> 256	1x1	Stride=1
layer1[0].bn3	BatchNorm2d	256	-	-
layer1[0].downsample	Conv2d + BatchNorm2d	64 -> 256	1x1	Stride=1
layer1[1]	Bottleneck	256 -> 256	-	No Downsampling
layer1[1].conv1	Conv2d	256 -> 64	1x1	Stride=1
layer1[1].bn1	BatchNorm2d	64	-	-
layer1[1].conv2	Conv2d	64 -> 64	3x3	Stride=1, Padding=1
layer1[1].bn2	BatchNorm2d	64	-	-
layer1[1].conv3	Conv2d	64 -> 256	1x1	Stride=1
layer1[1].bn3	BatchNorm2d	256	-	-
avgpool	AdaptiveAvgPool2d	-	1x1	Output=1x1
fc	Linear	2048 -> 6	-	Fully Connected Layer

Figure 3: ResNet_Model Architecture

Figure 3 highlights the significance of deep learning model for waste classification: ResNet-50, a deep convolutional neural network. It utilizes residual learning to obtain useful features from images in a way that efficient classification of different waste materials is possible. With its multi-layered architecture, raw waste images are translated into precise classifications.

IV. RESULTS

The performance captured through training and validation procedures shows the efficiency of the waste classification system using ResNet50. As clear from the performance table as well as the provided graphs, the model noted to increase in improvement throughout the duration of the 25 epochs of training [8][10].

At first, training as well as validation accuracy increased exponentially in the initial epochs. The validation accuracy was already over 97% by epoch 10 and then continued to exhibit steady performance thereafter. Both training and validation loss values decreased evenly, signifying that not only was the model learning well, but generalizing well to new data also [10].

Figure 4 shows accuracy comparison where both curves both increase and at some point plateau, signifying no overfitting. The loss graph shows a steady decrease, signifying convergence. All these conclusions confirm the

model's ability to identify visual difference between categories of waste with high accuracy [10].

These findings illustrate the applicability of the model to real-world applications, especially in settings where computational resources are limited. That high accuracy was attained without inordinate training cycles or specialized machinery further supports the model's worth for real-world waste classification [9][10].



Figure 4: Training vs. Validation Accuracy

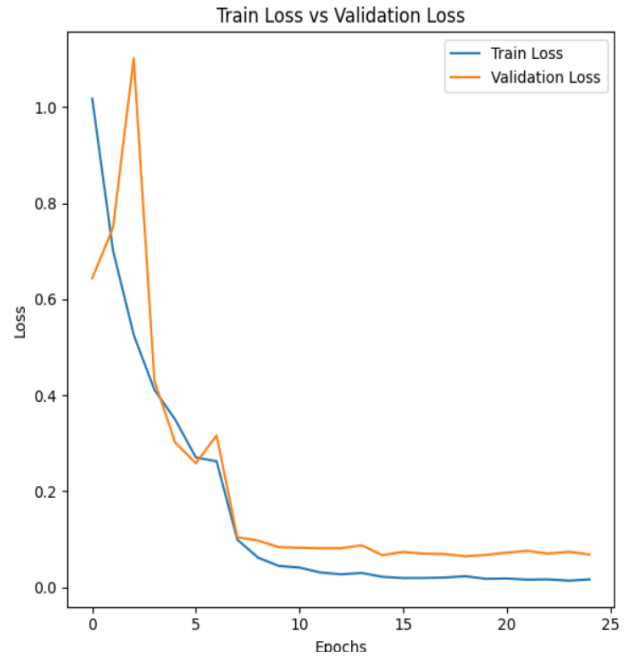


Figure 5: Training vs. Validation Loss

Figures 4 and 5 show the training and validation accuracy and loss over 25 epochs. The accuracy plot for training and validation (Figure. 4) shows that there is a uniform increase in the initial epochs, and by epoch 10, validation accuracy is

over 97.79%. After that, the plots begin to plateau, which shows that the model is able to distinguish the classes of waste satisfactorily without overfitting [8].

Further, the plot for training and validation loss (Figure. 5) also illustrates a steady fall during training. The smooth reduction of loss further validates that the model was continuing to improve predictions and building generalizable traits. The small distance between training and validation curves in both figures is an indication that the model generalised well and maintained consistent performance across various data [10].

Collectively, these plots verify that the model is precise and consistent and thus deployable to real waste sorting environments [6][9].

To test the trained model outside of graphs and metrics, a few actual test images were fed into the system. One of them is depicted in Figure 6, wherein the input image contained a promotional flyer composed of printed paper. The model accurately identified the label as "paper," demonstrating its ability to read material types well even in scenarios with lighting changes and text overlays [6], [8], [10]. This output confirms the appropriateness of the model for real-time deployment in waste identification applications [9].



Figure 6: Model output: successfully predicted "paper" class, confirming accurate real-time classification.

Predicted class: paper

Figure 7: Predicted output: classified as "paper"

V. CONCLUSION

The following paper investigated training and testing a waste classification model powered by an AI system with the ResNet50 deep neural network architecture. By using the TrashNet dataset, the system was trained on identifying six key types of waste under different visibilities. With the use of systematic preprocessing, efficient data augmentation, and orderly training, it was found that the system is highly capable to learn descriptive attributes and make viable predictions [5][6][7].

With a validation accuracy The present paper experimented with training and testing a waste classification model based on an AI system with the ResNet50 deep neural network architecture. Utilizing the TrashNet dataset, the system was trained to detect six major categories of waste in varying visibilities. With the application of systematic preprocessing, effective data augmentation, and organized training, it was discovered that the system is strongly capable to learn descriptive features and make feasible predictions.

With 97.79% validation accuracy and well-behaved training behavior in 25 epochs, the model clearly showed promise for practical application. The model proved resilient in relatively low computational resources, indicating its applicability in usage to smart waste management equipment [9][10].

By combining openly available software and a low-technology workflow, the project provides a low-cost, scalable solution to improving waste sorting. The model can be added to automated sorting machines, smart bins, or recycling centers reducing human intervention, increasing accuracy, and allowing for more sustainable cities. y of 97.79% and consistent training behavior over 25 epochs, the model clearly showed potential for real-world application [6][10].

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